Daily and Environmental Sound Classification Using Convolutional Neural Networks

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***Abstract*—A crucial problem in audio signal processing, environmental sound classification has applications in intelligent settings, smart cities, and surveillance systems. Environmental noises are difficult to recognize because, in contrast to speech, they are unstructured, non-stationary, and frequently tainted with background noise. Using the UrbanSound8K dataset, this work suggests a deep learning-based method for environmental sound classification. Convolutional Neural Network (CNN) architectures are used in parallel to extract Mel-Frequency Cepstral Coefficients (MFCCs) and Mel Spectrogram features from audio sources. To increase generalization, the model uses adaptive learning rate scheduling, batch normalization, dropout regularization, early stopping, and data augmentation. Accuracy, precision, recall, F1-score, and confusion matrix metrics were used to confirm the experimental results, which show strong categorization performance across ten urban sound categories.**

***Index Terms*—Environmental Sound Classification, Convolutional Neural Networks, MFCC, Deep Learning, UrbanSound8K**

I. Introduction

Because of its uses in autonomous systems, smart cities, surveillance systems, and healthcare monitoring, environmental sound recognition has grown in importance as a field of study. Environmental sounds are particularly difficult to classify because, in contrast to speech or music, they are usually unstructured, non-stationary, and frequently tainted by background noise.

Conventional machine learning techniques use manually created acoustic characteristics in conjunction with shallow classifiers like Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN). These techniques frequently struggle with complex and large-scale audio data, even though they can be useful for small-scale or well-controlled datasets.

By automatically learning discriminative representations from time-frequency audio information, recent developments in deep learning, particularly Convolutional Neural Networks (CNNs), have demonstrated impressive performance. Using Mel-Frequency Cepstral Coefficients (MFCCs) and Mel Spectrogram features taken from the UrbanSound8K dataset, this study investigates a CNN-based architecture for ambient sound classification. Data augmentation methods including pitch shifting, temporal stretching, and volume scaling are used to increase model robustness and generalization. In order to improve classification accuracy across various urban sound classes, the suggested method additionally integrates ensemble learning and parallel feature extraction.

II. Literature Review

Early studies on ambient sound categorization used classifiers like Gaussian Mixture Models (GMMs) and Support Vector Machines (SVMs) in conjunction with manually created features like Zero Crossing Rate (ZCR), spectral centroid, spectral roll-off, and MFCCs. Although these methods had some success, feature engineering was crucial to their effectiveness.

A. MFCC Features

One of the most used characteristics in speech and audio processing is Mel-Frequency Cepstral Coefficients, or MFCCs. By mapping frequencies onto the Mel scale and using a logarithmic compression, MFCCs are intended to resemble human auditory perception. MFCCs are useful for sound classification problems with low processing complexity because the generated cepstral coefficients offer a concise representation of the spectral envelope.

B. Mel Spectrogram Features

The short-time power spectrum of an audio stream projected onto the Mel scale is represented by a Mel spectrogram. Mel spectrograms are ideal for convolutional neural networks because they retain more precise time-frequency information and spatial localization than MFCCs. According to recent research, CNNs trained on mel spectrograms frequently perform better than conventional feature-based methods because they can immediately learn hierarchical spectral patterns from the input.

Using log-mel spectrograms for environmental sound classification, Piczak [**?**] showed how effective CNNs are. The UrbanSound8K dataset was presented by Salamon et al. [**?**], who also emphasized the significance of data augmentation methods including time stretching and pitch shifting. In order to increase training stability and generalization, more recent research emphasizes the use of batch normalization, dropout, and adaptive learning rate scheduling.

Inspired by these works, this project integrates MFCC features with a CNN architecture enhanced by regularization and augmentation techniques.

III. Comparison with Previous Methods

Table I summarizes some recent studies on environmental sound classification and compares them with the proposed CNN-based approach using MFCC and Mel Spectrogram features.

TABLE I
Comparison of Previous Methods and Proposed Approach

Method	Features	Model	Acc. (%)
Tzanetakis & Cook (2002)	MFCC	k-NN/SVM	64.0
Piczak (2015)	Log-Mel	CNN	73.7
Salamon et al. (2017)	Log-Mel + Aug	CNN	79.1
Proposed MFCC-CNN	MFCC	CNN	65.8
Proposed Mel-Spec-CNN	Mel Spec	CNN	71.1

IV. Dataset Description

The UrbanSound8K dataset, which consists of 8,732 brief audio recordings with a maximum duration of four seconds, is a commonly used benchmark for ambient sound classification. Ten different urban sound classes—air conditioner, vehicle horn, kids playing, dog bark, drilling, engine idling, gunshot, jackhammer, siren, and street music—are covered by the dataset.

To make cross-validation easier, the audio snippets are arranged into ten stratified folds. Folds 1 through 8 are used for training, fold 9 for validation, and fold 10 for testing in this investigation. By avoiding data leaks between the training, validation, and test sets, this partitioning technique guarantees an objective assessment of the model.

V. Feature Extraction

A. MFCC-Based Feature Extraction

The first experimental setup uses 40 coefficients with a frame size of 1024 samples and a hop length of 512 samples to extract MFCC characteristics from each audio source. To guarantee a consistent input size for the CNN model, the MFCC matrices are padded or trimmed to a fixed dimension of 40×174.

B. Mel-Spectrogram-Based Feature Extraction

In the second configuration, 128 Mel bands are used to calculate log-scaled mel spectrograms. To keep the spectrograms compatible with the CNN architecture, they are shrunk to a fixed dimension and translated to decibel scale. Compared to MFCC characteristics, this approach preserves richer spectral and temporal information.

VI. Data Augmentation

During the training phase, data augmentation techniques are used to increase model robustness and reduce overfitting. In particular, random changes including time stretching, pitch shifting, and loudness scaling are applied to the audio data. While maintaining the basic class features of the sound signals, these augmentations add realistic variances. The model performs better on the validation and test sets because it can more easily generalize to new data by simulating a variety of acoustic circumstances.

VII. Proposed CNN Architecture

A. CNN for MFCC Features

A lightweight convolutional neural network (CNN) architecture is used to effectively process small spectral data for MFCC-based classification. To minimize spatial dimensions while maintaining discriminative information, the network comprises of two convolutional layers with tiny 3×3 kernels, each followed by max pooling and ReLU activation.

The CNN is able to identify local time-frequency patterns since the extracted MFCC features are handled as two-dimensional inputs. To reduce overfitting, especially because of the relatively small feature dimensionality, a dropout layer is included before the fully connected layers. Multi-class ambient sound recognition is made possible by the use of fully connected layers with a softmax output for the final classification. A good trade-off between computing efficiency and classification accuracy is provided by this architecture.

B. CNN for Mel Spectrogram Features

A deeper CNN architecture is used for mel spectrogram-based classification in order to take use of the input data's higher time-frequency representation. Three convolutional layers with progressively deeper filters make up the model, which enables hierarchical feature extraction from low-level spectral edges to higher-level auditory patterns. To increase training stability and convergence, batch normalization and ReLU activation come after each convolutional layer.

Dropout regularization is utilized to improve generalization performance, while max pooling layers are applied to gradually decrease spatial resolution. The deeper architecture allows for more efficient learning of complicated sound features since mel spectrograms have a better resolution and a more sophisticated structure. The learnt representations are mapped to class probabilities by the final fully connected layers, which improves discrimination between acoustically identical sound classes.

VIII. Training Methodology

A supervised learning method using categorical cross-entropy as the objective function is used to train the suggested models. With an initial learning rate of 0.001 and L2 regularization using weight decay to reduce overfitting, the Adam optimizer is used because of its flexible learning rate capacity. With a batch size of 32, training is carried out for a maximum of 30 epochs.

An adaptive learning rate scheduler that lowers the learning rate when validation accuracy reaches a plateau is used to improve convergence stability and generalization performance. In order to avoid needless computation and overfitting, early stopping is used to cease training if no improvement in validation performance is seen for five consecutive epochs. For the final assessment on the test set, the model with the best validation accuracy is kept.

Separate training experiments are conducted for MFCC and mel spectrogram features, with CNN architectures tailored to the respective feature representations. This enables a fair

and systematic comparison of both approaches under identical training conditions.

A. System Flowchart

Figure 1 illustrates the overall workflow of the proposed environmental sound classification system. The pipeline begins with raw audio acquisition, followed by preprocessing and feature extraction. Depending on the experimental setup, MFCC or mel spectrogram features are extracted and passed to the corresponding CNN architecture for training and evaluation, ultimately producing the predicted sound class.

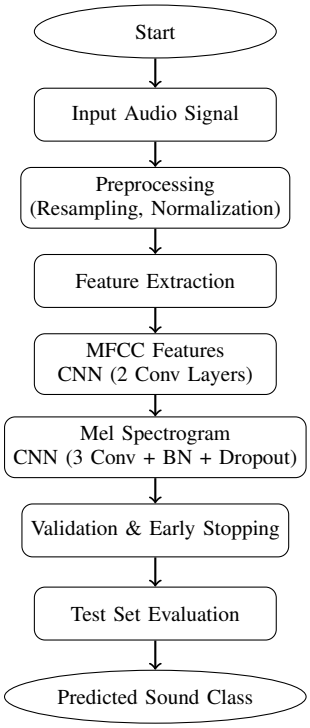

Fig. 1. Comparative workflow for MFCC and Mel spectrogram-based environmental sound classification

B. Feature-Based Processing Flow

Figure 1 illustrates the comparative processing pipelines for MFCC and Mel spectrogram-based classification. While both approaches share common preprocessing stages, they differ in feature representation and CNN architecture depth.

C. Hyperparameter Configuration

Table II summarizes the key hyperparameters used for training both MFCC-based and mel spectrogram-based CNN models.

TABLE II
HYPERPARAMETER CONFIGURATION

Parameter	Value
Dataset	UrbanSound8K
Sampling Rate	22,050 Hz
Feature Type	MFCC
Number of MFCCs	40
Frame Size (FFT)	1024
Hop Length	512
Input Dimension	40×174
Batch Size	32
Optimizer	Adam
Initial Learning Rate	0.001
Weight Decay	1×10^{-3}
Loss Function	Cross-Entropy Loss
Dropout Rate	0.5
Maximum Epochs	30
Learning Rate Scheduler	ReduceLROnPlateau
Early Stopping Patience	5 epochs

All hyperparameters are kept consistent across experiments to ensure a fair comparison between MFCC and mel spectrogram feature representations.

D. Hardware and Training Details

All experiments were conducted using Google Colab with the following hardware configuration:

- GPU: T4
- CPU: AMD Ryzen
- RAM: 12 GB

The training process for both MFCC and Mel Spectrogram-based CNNs consisted of 30 epochs with a batch size of 32. On this setup, the total training time was approximately 2.5 hours, averaging about 5 minutes per epoch for the Mel Spectrogram-based CNN and slightly less for the MFCC-based CNN due to smaller input dimensions.

IX. EXPERIMENTAL RESULTS

Accuracy, precision, recall, and F1-score are used to assess the model's performance on the test set. Additionally, a confusion matrix is created to examine performance by class. Loss and accuracy curves for training and validation show steady convergence and efficient regularization.

A. MFCC-Based Results

On the test set, the MFCC-based CNN shows competitive classification performance and consistent convergence during training. The model has a quicker training time since MFCC features are compact, but it has trouble differentiating between acoustically identical sound classes.

1) Training Curves: Figure 2 and 3 illustrates the training and validation loss and accuracy for the MFCC-based CNN model.

2) Confusion Matrix: The confusion matrix shown in Figure 4 summarizes the class-wise performance of the MFCC-based model on the test set.

Fig. 2. Training and validation loss curves for MFCC-based CNN

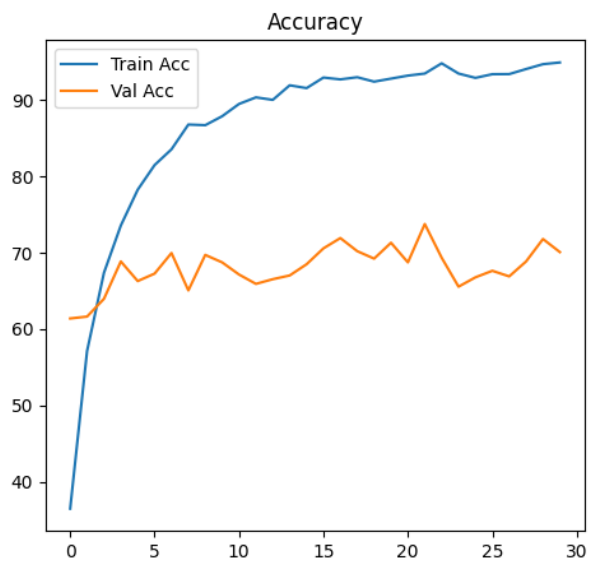

Fig. 3. Training and validation accuracy curves for MFCC-based CNN

B. Mel-Spectrogram-Based Results

Higher overall accuracy and better class-wise discrimination are attained by the CNN trained on mel spectrogram features. Although it comes at the expense of more computer complexity, the richer time-frequency representation allows for greater separation of complicated environmental noises.

1) Training Curves: Figure 5 6 and presents the training dynamics of the mel spectrogram-based CNN.

2) Confusion Matrix: Figure 7 shows the confusion matrix obtained on the test dataset.

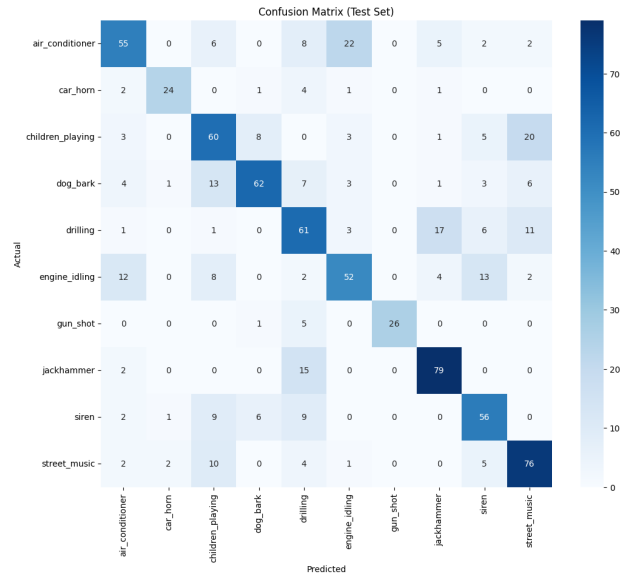

Fig. 4. Confusion matrix for MFCC-based CNN

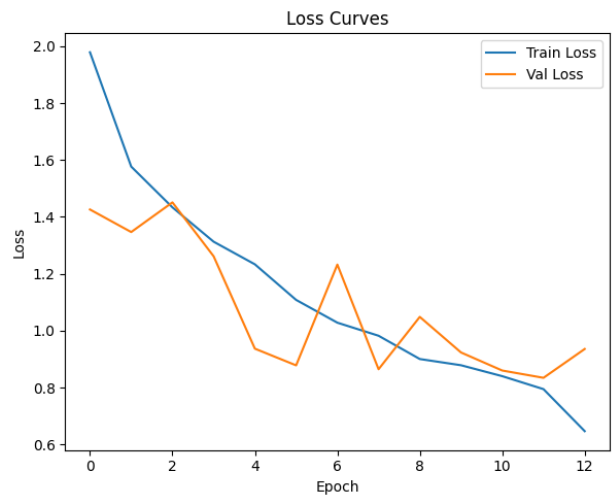

Fig. 5. Training and validation loss curves for mel spectrogram-based CNN

C. Comparative Performance Analysis

TABLE III
COMPARISON OF MFCC AND MEL-SPECTROGRAM-BASED MODELS

Metric	MFCC-CNN	Mel-Spec-CNN
Training Accuracy (%)	94.96	78.63
Validation Accuracy (%)	73.77	76.72
Testing Accuracy (%)	65.83	71.09
Precision (%)	66.77	72.03
Recall (%)	65.83	71.09
F1-score (%)	65.90	71.08

From Table III, it is evident that the CNN based on mel-spectrograms consistently performs better than the MFCC-based model. Due to the compact and lossy nature of MFCC features, the MFCC-CNN's lower testing accuracy suggests

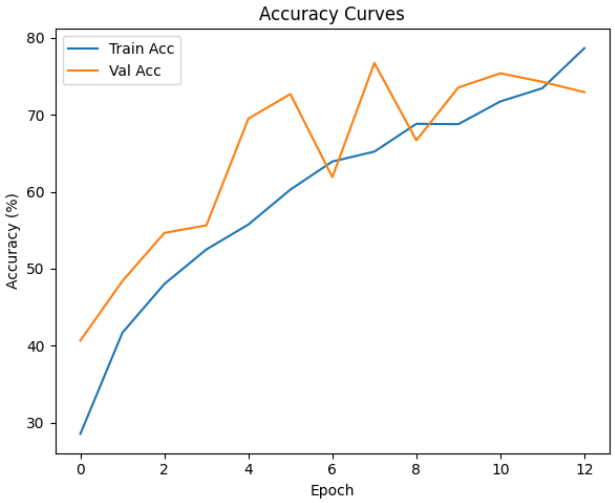

Fig. 6. Training and validation accuracy curves for mel spectrogram-based CNN

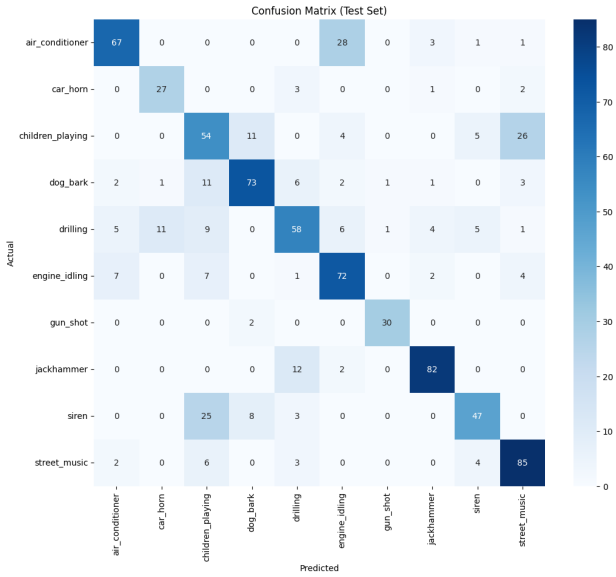

Fig. 7. Confusion matrix for mel spectrogram-based CNN

overfitting even while it achieves better training accuracy. Mel spectrograms, on the other hand, preserve richer time-frequency information, allowing for stronger class-wise classification and generalization. This performance trade-off demonstrates that whereas mel spectrograms provide greater accuracy at the expense of greater computational complexity, MFCC features are computationally economical but less expressive.

X. Analysis and Discussion

Mel spectrogram-based models regularly outperform MFCC-based models across all assessment measures, according to the comparative analysis. CNNs are better able to learn spatially confined spectrum patterns because of the richer time-frequency resolution retained in mel spectrogram representations.

However, MFCC-based models offer lower computational complexity and faster training, making them appropriate for resource-constrained applications. The trade-off between efficiency and accuracy emphasizes how crucial it is to choose features according to application needs.

XI. Limitations and Future Work

The suggested models have some drawbacks even though they achieve competitive classification performance. The fixed-size feature representations used by the MFCC-based CNN may eliminate fine-grained temporal dynamics and restrict the ability to distinguish between acoustically identical sound classes. The CNN based on mel-spectrograms enhances generalization, but it requires more training time and computer complexity.

Additionally, while both models use convolutional architectures to capture local time-frequency patterns, they do not explicitly model long-term temporal dependencies. In order to better capture temporal context, future research may investigate transformer or attention-based designs. To enable real-time deployment on systems with limited resources, further enhancements can involve experimenting with raw waveform-based learning, sophisticated data augmentation methods, and model compression approaches.

XII. Conclusion

This project demonstrated a comparative deep learning-based system for classifying everyday and ambient sounds using convolutional neural networks with MFCC and mel spectrogra m features. Experimental results on the UrbanSound8K dataset show that the mel-spectrogram-based CNN achieves better classification accuracy and greater generalization, while the MFCC-based CNN offers faster training and reduced computing cost.

The study emphasizes the trade-off between representational richness and computing efficiency when choosing audio features. The suggested approach provides strong performance across a variety of ambient sound classes using efficient feature extraction, data augmentation, and adaptive training techniques. To improve accuracy and practicality, future developments might include lightweight structures, transformer-based models, or attention techniques.

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